

PHYS 143 – Mechanics and Thermal Physics
Week 6: Gravitation – Tutorial Sheet

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Use the Master Procedure: identify, sketch (mark the centre and r), predict, choose the tool (Kepler ratio, force, circular orbit, energy), execute, evaluate. Constants: $G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$, $M_E = 5.972 \times 10^{24} \text{ kg}$, $R_E = 6.371 \times 10^6 \text{ m}$, $GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, $M_\odot = 1.989 \times 10^{30} \text{ kg}$, $GM_\odot = 1.327 \times 10^{20} \text{ m}^3/\text{s}^2$, $1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$, $g_s = 9.82 \text{ m/s}^2$. Section A holds the worked examples moved out of the lecture; final answers to B, C, D and E are at the end, and the exam-style page has its own answers.

A. Worked examples (extra to the lecture)

A1. The Earth's speed at perihelion and aphelion

Earth: $r_p = 1.471 \times 10^{11} \text{ m}$, $r_a = 1.521 \times 10^{11} \text{ m}$, average speed 29.8 km/s . Find v_p and v_a .

Solution. Kepler 2 at the two ends: $v_p r_p = v_a r_a$, so $v_p/v_a = r_a/r_p = 1.034$. With $\frac{1}{2}(v_p + v_a) \approx 29.8$: $v_a(1 + 1.034)/2 = 29.8$, so $v_a = 29.3 \text{ km/s}$ and $v_p = 30.3 \text{ km/s}$. Check: $30.3 > 29.8 > 29.3$, a 3.4% spread, as the near-circular orbit predicts.

A2. Weight at the equator

A 70.0 kg person stands on a scale at the equator. Find F_g , the reading n , and the fractional difference. (Sidereal day $86\,164 \text{ s}$.)

Solution. $\omega = 2\pi/86\,164 = 7.292 \times 10^{-5} \text{ rad/s}$, $\omega^2 R_E = 0.0339 \text{ m/s}^2$. $F_g = 70.0(9.82) = 687 \text{ N}$ and $n = F_g - m\omega^2 R_E = 687 - 70.0(0.0339) = 685 \text{ N}$: 0.35% smaller. At the poles $n = F_g$; at latitude 45° the effect is smaller because the circle is smaller.

A3. Weighing the Sun and the Earth

(a) The Earth orbits the Sun at $r = 1.496 \times 10^{11} \text{ m}$ with $T = 3.156 \times 10^7 \text{ s}$. Find $M_\odot$. (b) The Moon orbits at $3.844 \times 10^8 \text{ m}$ with $T = 27.32 \text{ days}$. Find M_E .

Solution. Circular orbit, $T^2 = 4\pi^2 r^3/GM$, so $M = 4\pi^2 r^3/(GT^2)$. (a)

$$M_\odot = \frac{4\pi^2(1.496 \times 10^{11})^3}{(6.674 \times 10^{-11})(3.156 \times 10^7)^2} = 1.99 \times 10^{30} \text{ kg}$$

(b) $T = 27.32(86\,400) = 2.360 \times 10^6 \text{ s}$, so $M_E = 6.03 \times 10^{24} \text{ kg}$ (within 1%; the Moon's own mass is neglected).

A4. Escape from the Moon and from Mars

Moon: $M = 7.35 \times 10^{22} \text{ kg}$, $R = 1.737 \times 10^6 \text{ m}$. Mars: $M = 6.417 \times 10^{23} \text{ kg}$, $R = 3.390 \times 10^6 \text{ m}$. Find v_{esc} for each.

Solution. $v_{\text{esc}} = \sqrt{2GM/R}$. Moon: $\sqrt{2(6.674 \times 10^{-11})(7.35 \times 10^{22})/1.737 \times 10^6} = 2.38 \text{ km/s}$. Mars: 5.03 km/s . Both lie below the Earth's 11.2 km/s , in the order of their M/R .

A5. Raising a satellite into orbit

A 1000 kg satellite is raised from rest on the ground to a circular orbit 1000 km up. Find the energy that must be supplied (ignore rotation and air).

Solution. $r = 7.371 \times 10^6$ m.

$$\begin{aligned}\Delta E &= E_{\text{orbit}} - E_{\text{ground}} = GMm \left(\frac{1}{R_E} - \frac{1}{2r} \right) \\ &= 3.986 \times 10^{17} (1.570 \times 10^{-7} - 6.783 \times 10^{-8}) = 3.55 \times 10^{10} \text{ J}\end{aligned}$$

Of this $K_{\text{orbit}} = 2.70 \times 10^{10}$ J and $\Delta U = 0.85 \times 10^{10}$ J: most of the energy gives orbital speed, not height.

A6. Halley's comet: speeds at both ends

$a = 2.668 \times 10^{12}$ m, $e = 0.967$. Find v_p and v_a .

Solution. $r_p = a(1 - e) = 8.80 \times 10^{10}$ m, $r_a = a(1 + e) = 5.25 \times 10^{12}$ m. Vis-viva: $v_p = \sqrt{GM_{\odot}(2/r_p - 1/a)} = 5.45 \times 10^4$ m/s. Then $v_a = v_p r_p / r_a = 9.1 \times 10^2$ m/s. Check $v_p r_p = 4.79 \times 10^{15} = v_a r_a$, and $T = 2\pi\sqrt{a^3/GM_{\odot}} = 75.3$ years.

A7. Jupiter from one moon

Io orbits Jupiter at $a = 4.217 \times 10^8$ m with $T = 1.769$ days. Jupiter's radius is 7.149×10^7 m. Find the speed needed to leave Jupiter's gravity from its cloud tops.

Solution. $M_J = \frac{4\pi^2 a^3}{GT^2} = \frac{4\pi^2 (4.217 \times 10^8)^3}{(6.674 \times 10^{-11})(1.528 \times 10^5)^2} = 1.90 \times 10^{27}$ kg, then $v_{\text{esc}} = \sqrt{2GM_J/R_J} = 5.95 \times 10^4$ m/s.

A8. An integrated problem: the method is not given

A 2.0×10^3 kg spacecraft is in a circular orbit 300 km above the Earth. (a) Find its speed and period. (b) How much additional speed, along its motion, is needed to escape? (c) Find its total energy and the energy to be added to reach $E = 0$.

Solution. $r = 6.671 \times 10^6$ m. (a) $v = \sqrt{GM/r} = 7.73$ km/s, $T = 2\pi r/v = 90.4$ min. (b) $v_{\text{esc}} = \sqrt{2}v = 10.93$ km/s, so the extra speed is 3.20 km/s. (c) $E = -GMm/2r = -5.97 \times 10^{10}$ J; adding $+5.97 \times 10^{10}$ J (equal to K) gives $E = 0$.

B. Guided problems (hints given)

B1. Mercury

Mercury has $a = 0.387$ AU and $e = 0.206$. Find r_p and r_a (in AU), and its period in years using Earth's orbit as the reference. *Hint: Toolkits 1 and 2.*

B2. A comet

A comet has $r_p = 0.50$ AU and $r_a = 4.5$ AU. Find a , e , its period, and v_p/v_a . *Hint: $a = (r_p + r_a)/2$; Kepler 3 in years and AU.*

B3. A higher satellite

A satellite is in a circular orbit of radius $2R_E$. Find its speed and period, and its acceleration. *Hint: r is the radius, so the altitude is R_E .*

B4. The force between the Earth and the Moon

Find the gravitational force between them ($M_{\text{Moon}} = 7.348 \times 10^{22} \text{ kg}$, separation $3.844 \times 10^8 \text{ m}$). *Hint: Newton's third law.*

B5. The null point

Where on the line from the Earth to the Moon does the net gravitational force on a small spacecraft vanish? *Hint: set the two magnitudes equal, $M_E/x^2 = M_M/(d-x)^2$.*

C. Independent problems

C1. Three equal masses

Three 5.0 kg masses sit at the corners of an equilateral triangle of side 0.20 m . Find the net gravitational force on one of them, and its direction.

C2. A moon of Mars

Phobos orbits Mars at $a = 9.38 \times 10^6 \text{ m}$ with $T = 7.65 \text{ hours}$. Find the mass of Mars.

C3. Falling from far away

A meteoroid is at rest at $r = 3R_E$ (centre of the Earth). With what speed does it reach the surface (ignore air)?

C4. Drag

A satellite in a 400 km orbit loses energy and settles in a circular orbit at 390 km . Find the change in its speed and in its energy per kilogram. Does it speed up?

C5. Escape from orbit

Find the escape speed of a spacecraft at an altitude of 400 km , and compare with the speed in a circular orbit there.

C6. An elliptical orbit

A satellite has perigee altitude 300 km and apogee altitude 3000 km . Find a , e , the period, and its speeds at perigee and apogee. *Remember r is from the centre.*

D. Application problems

D1. Does the apple fall like the Moon?

Compute the Moon's centripetal acceleration from GM_E/r^2 at $r = 3.844 \times 10^8$ m and compare with $g_s/60^2$. What do you conclude about the law of gravitation?

D2. Pluto

Pluto has $a = 5.91 \times 10^{12}$ m and $e = 0.249$. Find its closest and farthest distances from the Sun, and the ratio of its speeds there.

D3. A two-hour orbit

A mapping satellite should orbit the Earth once every 2.00 hours. Find its altitude and speed.

D4. Challenge: the method is not given

A probe is to be launched from the Earth's surface so that it just reaches the altitude of the geostationary orbit ($r = 4.216 \times 10^7$ m) and stops there momentarily. Find the launch speed (ignore rotation and air). Is it above or below the escape speed?

E. Concept questions

1. The Sun is at a focus of Earth's orbit. Is the Earth's speed greater in January or July? Why?
2. Why does the mass of the satellite not appear in the orbital period?
3. An astronaut is "weightless" in the ISS. Is gravity zero there? Explain with g and weight.
4. Why is $U = mgh$ wrong for a launch to $h = R_E$?
5. A satellite in a circular orbit loses a little energy to drag. Does it speed up or slow down? Explain with $E = -GMm/2r$.
6. Does the escape speed depend on the direction of launch, or on the mass of the projectile?

Answers to B to E

- B1 $r_p = 0.307 \text{ AU}$, $r_a = 0.467 \text{ AU}$; $T = 0.387^{3/2} = 0.241 \text{ yr}$
- B2 $a = 2.5 \text{ AU}$, $e = 0.80$, $T = 2.5^{3/2} = 3.95 \text{ yr}$, $v_p/v_a = r_a/r_p = 9$
- B3 $v = \sqrt{GM/2R_E} = 5.59 \text{ km/s}$, $T = 3.98 \text{ h}$, $a = g_s/4 = 2.45 \text{ m/s}^2$
- B4 $1.98 \times 10^{20} \text{ N}$ on each, in opposite directions
- B5 $x = d/(1 + \sqrt{M_M/M_E}) = 3.46 \times 10^8 \text{ m}$ from the Earth ($0.900 d$)
- C1 each pair $F = G(5.0)^2/(0.20)^2 = 4.17 \times 10^{-8} \text{ N}$; two at 60° : net $2F \cos 30^\circ = 7.22 \times 10^{-8} \text{ N}$, toward the centre of the triangle
- C2 $M = 4\pi^2 a^3/(GT^2) = 6.44 \times 10^{23} \text{ kg}$
- C3 $v = \sqrt{2GM(1/R_E - 1/3R_E)} = 9.13 \text{ km/s}$
- C4 v : $7.672 \rightarrow 7.678 \text{ km/s}$ ($+5.7 \text{ m/s}$, it speeds up); $\Delta E = -4.35 \times 10^4 \text{ J/kg}$ (more negative)
- C5 $v_{\text{esc}} = 10.85 \text{ km/s}$; circular 7.67 km/s ; ratio $\sqrt{2}$
- C6 $r_p = 6.671 \times 10^6 \text{ m}$, $r_a = 9.371 \times 10^6 \text{ m}$; $a = 8.021 \times 10^6 \text{ m}$; $e = 0.168$; $T = 119.2 \text{ min}$; $v_p = 8.35 \text{ km/s}$, $v_a = 5.95 \text{ km/s}$ ($v_p r_p = v_a r_a$ checks)
- D1 $a = GM/r^2 = 2.70 \times 10^{-3} \text{ m/s}^2$; $g_s/60^2 = 2.73 \times 10^{-3}$ (the Moon is $60.3R_E$ away): agreement to 1% supports the inverse-square law
- D2 $r_p = 4.44 \times 10^{12} \text{ m}$, $r_a = 7.38 \times 10^{12} \text{ m}$; $v_p/v_a = r_a/r_p = 1.66$
- D3 $r = 8.06 \times 10^6 \text{ m}$, altitude $1.69 \times 10^3 \text{ km}$; $v = 7.03 \text{ km/s}$
- D4 energy: $\frac{1}{2}v_0^2 = GM(1/R_E - 1/r)$: $v_0 = 10.3 \text{ km/s}$ (with $1/R_E = 1.570 \times 10^{-7}$, $1/r = 2.372 \times 10^{-8} \text{ m}^{-1}$), below 11.2 km/s
- E1 January (perihelion, $v_p r_p = v_a r_a$)
- E2 m cancels between GMm/r^2 and mv^2/r
- E3 $g \approx 8.7 \text{ m/s}^2$ there; weight (the normal force) is zero because astronaut and station are in free fall together
- E4 g falls by a factor 4 over the climb; mgh is 62.6 against 31.3 MJ/kg
- E5 speeds up: E falls, r falls, $v = \sqrt{GM/r}$ rises ($K = -E$)
- E6 neither (the energy argument gives $\sqrt{2GM/R}$; the path must clear the planet)

F. Exam-style page (ten five-option items, about 20 minutes)

Circle one answer. Use $GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, $R_E = 6.371 \times 10^6 \text{ m}$, $g_s = 9.82 \text{ m/s}^2$.

- According to Kepler's laws,
 - a planet moves fastest when it is farthest from the Sun;
 - the line joining a planet to the Sun sweeps equal areas in equal times;
 - the square of the period is proportional to the cube of the semi-major axis.(A) I only (B) II only (C) III only (D) I and II (E) II and III
- A planet's orbit has $r_p = 3.0 \times 10^{11} \text{ m}$ and $r_a = 5.0 \times 10^{11} \text{ m}$. Its eccentricity is
(A) 0.20 (B) 0.25 (C) 0.40 (D) 0.60 (E) 0.75
- A planet has semi-major axis $2.0 \times 10^{11} \text{ m}$ and eccentricity 0.30. Its closest distance to the Sun is
(A) $0.60 \times 10^{11} \text{ m}$ (B) $1.4 \times 10^{11} \text{ m}$ (C) $1.7 \times 10^{11} \text{ m}$ (D) $2.6 \times 10^{11} \text{ m}$ (E) $3.0 \times 10^{11} \text{ m}$
- Satellite 1 orbits the Earth at radius R and satellite 2, of the same mass, at $2R$. The period of satellite 2 divided by that of satellite 1 is
(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4 (E) 8
- A satellite is 300 km above the Earth and its radial acceleration is 8.96 m/s^2 . Its orbital speed is
(A) 5.50 km/s (B) 6.62 km/s (C) 7.73 km/s (D) 8.84 km/s (E) 9.95 km/s
- At an altitude equal to the Earth's radius, the acceleration due to gravity is
(A) 9.82 m/s^2 (B) 4.91 m/s^2 (C) 3.27 m/s^2 (D) 2.45 m/s^2 (E) 1.09 m/s^2
- About a satellite in a circular orbit,
 - its period depends on the satellite's mass;
 - if drag lowers its orbit, its speed increases;
 - the astronauts inside are weightless because gravity is zero there.(A) I only (B) II only (C) III only (D) I and III (E) II and III
- A planet has twice the Earth's mass and the same radius. Its escape speed is about
(A) 5.6 km/s (B) 11.2 km/s (C) 15.8 km/s (D) 22.4 km/s (E) 31.7 km/s
- A satellite in a circular orbit has kinetic energy K . Its total energy (zero at infinite separation) is
(A) $-2K$ (B) $-K$ (C) $-K/2$ (D) 0 (E) $+K$
- Which of the statements is/are correct?
 - The gravitational potential energy $-GMm/r$ is negative at every finite r .
 - The escape speed from a planet depends on the mass of the projectile.
 - A body with total energy $E < 0$ is bound to the planet.(A) I only (B) III only (C) I and III only (D) II and III only (E) I, II and III

Answers. 1 (E): I is false (fastest at perihelion). 2 (B): $e = (r_a - r_p)/(r_a + r_p) = 2.0/8.0 = 0.25$. 3 (B): $r_p = a(1 - e) = 1.4 \times 10^{11} \text{ m}$. 4 (C): $T \propto r^{3/2}$, $2^{3/2} = 2\sqrt{2}$. 5 (C): $r = 6.671 \times 10^6 \text{ m}$, $v = \sqrt{a_r r} = 7.73 \text{ km/s}$. 6 (D): $r = 2R_E$, $g = g_s/4 = 2.45 \text{ m/s}^2$. 7 (B): II only (period independent of m ; drag speeds the satellite up; weightless because of free fall, not zero gravity). 8 (C): $\sqrt{2} \times 11.2 = 15.8 \text{ km/s}$. 9 (B): $E = -K$. 10 (C): II is false (m cancels), I and III are true.